Draft Project Report: Open-Source NGFW with Adaptive Detection

1. Introduction

This project studies how an open-source firewall setup can be extended toward next-generation firewall behavior using data-driven detection and feedback. The implementation uses OPNsense as inline firewall, Suricata for IDS/IPS, and a side analytics node for telemetry processing, anomaly scoring, and reporting.

In simple words, the project tries to close this gap:

1. normal rule/signature based open-source firewalling,
2. adaptive behavior-aware detection mostly available in commercial NGFW products.

The project is in network security domain, and focuses on practical deployment that can actually be reproduced in a humble lab.

1.1 Scope of Project

The scope includes architecture, implementation, and evaluation of a sidecar model with these parts:

1. stateful segmentation and policy enforcement with OPNsense,
2. IDS/IPS detection through Suricata,
3. log collection and visualization,
4. anomaly and classification pipeline using Isolation Forest and Random Forest,
5. controlled traffic generation for testing using scenario scripts and TRex.

1.2 Scope of Project Impact

1.2.1 Social Impact

1. Gives a usable cybersecurity demo platform for researchers and students.
2. Helps understand real monitoring and attack-defense correlation, not just theory.
3. Encourages safer network design habits like segmentation and least privilege.

1.2.2 Economic Impact

1. Uses open-source stack and avoids commercial NGFW licensing.
2. Can be replicated by small colleges or labs with limited budgets.
3. Skills learned map to SOC and network security industry roles.

1.2.3 Environmental Impact

1. Reuses existing hardware through virtualization instead of extra appliances.
2. Modular design allows only required components to run, reducing unnecessary overhead.
3. High-volume experiments still consume power, so workload profiling is required.

1.2.4 Academic Impact

1. Connects literature to implementation.
2. Produces reusable artifacts: configs, scripts, dataset snapshots, and results.
3. Sets a base for future work on closed-loop response.

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2. Problem Statement

Most open-source firewall deployments are strong in packet filtering and signature matching, but not adaptive by default. They depend on pre-defined rules and known signatures. That makes them weak against behavior-level anomalies, zero-day style events, and dynamic attack patterns.

Main practical issues are:

1. manual dependence for tuning and response,
2. delayed reaction to unknown threats,
3. disconnected pipeline between detection and enforcement,
4. difficulty in producing local environment-specific intelligence.

So the core project question is:

How to augment an open-source firewall stack with lightweight adaptive detection and telemetry-driven feedback, while keeping it transparent, low-cost, and implementable in an academic setup?

<image of current open-source firewall workflow describing where static rules/signatures stop being enough for adaptive threat handling>

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3. Literature Review

This section summarizes the findings from the survey literature and how they informed this project design.

3.1 Firewall Evolution Summary

Literature tracks evolution from packet filters to proxy firewalls, then stateful inspection, UTM, and NGFW. NIST guidance and later NGFW surveys show that each phase added visibility and control depth.

Key timeline used in this report:

Year	Milestone
1988	Morris Worm incident and CERT/CC formation
1989	Early packet filtering (`screend`)
1991	Proxy firewall deployment pattern
1994	Stateful inspection mainstreamed (FireWall-1 era)
1998	Snort released
2004	UTM market category formalized
2009	NGFW category formalized
2015	OPNsense fork introduced
2020s	Adaptive detection research grows

<image of firewall evolution flow showing packet filter -> proxy -> stateful -> UTM -> NGFW -> adaptive firewall>

3.2 NGFW Capability from Literature

Neupane et al., Liang and Kim, and Heino et al. emphasize that NGFW value is not only feature count, but integration quality: application context, DPI, threat intelligence, and prevention in one operational path.

Component	Typical Layer Coverage	Function
Packet filtering	L3/L4	ACL based allow/deny
Stateful engine	L4	Session tracking
DPI	L4-L7	Payload/protocol inspection
App identification	L7	Application-aware control
IDS/IPS	L3-L7	Threat detection and blocking
Threat intelligence	L3-L7	IoC correlation

3.3 IDS and Adaptive Detection Literature

Denning (1987) established anomaly-based IDS concepts. Khraisat et al. and Liao et al. discuss operational challenges like false positives and alert overload. Ahmad et al. and Dini et al. show there is still a gap between research models and production security operations.

3.4 Isolation Forest Motivation from Surveyed Papers

Liu et al. introduced Isolation Forest as an unsupervised anomaly method. Tao et al. validated it in network traffic settings, and Al Farizi et al. reported good efficiency across studies. For this project, that is useful because lab data labeling is not always complete.

3.5 Identified Gap in Open-Source Ecosystem

From the surveyed papers, the recurring gap is this: open-source firewall stacks are mature in static controls but weak in adaptive behavior-level learning and automatic policy feedback.

System Class	Detection Style	Inline Prevention	Adaptability	Open-Source Availability
Stateless firewall	Rule only	Yes	Low	High
Stateful firewall	Rule + state	Yes	Low	High
IDS	Signature/anomaly alerting	No	Low	High
IPS	Signature/anomaly	Yes	Low	High
UTM	Multi-engine	Yes	Moderate	Partial
NGFW	Context + signature	Yes	Moderate	Limited
Adaptive sidecar firewall	Learned behavior + signatures	Yes	High	Underexplored

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4. Project Description

The project implements a sidecar-based adaptive firewall architecture.

4.1 System Overview

1. OPNsense is the inline enforcement point.
2. Suricata performs signature-driven IDS/IPS detection.
3. Firewall and IDS events are exported using syslog.
4. Analytics node processes logs into features.
5. Isolation Forest gives anomaly score trends.
6. Random Forest classifies known scenario classes.
7. Dashboard combines control-plane and data-plane visibility.

<image of sidecar architecture showing enforcement on firewall and analytics/inference on separate node>

4.2 Main Use Cases

1. Student cyber range and practical coursework.
2. Security monitoring training and triage exercises.
3. Comparative study: signature-only vs adaptive assisted detection.
4. Controlled segmentation validation under mixed traffic.

4.3 Importance of This Project

1. Shows that adaptive firewall behavior can be approximated with open tooling.
2. Makes NGFW-like experimentation possible in low-resource settings.
3. Produces a practical workflow that can be reused by later batches.

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5. Requirements

5.1 Functional Requirements

1. Enforce segmented policy between red and blue networks.
2. Detect suspicious activity through Suricata rules.
3. Export logs to centralized analytics stack.
4. Extract and store network features for model training and scoring.
5. Generate anomaly and class outputs and present them in dashboard.
6. Support repeatable scenario-based traffic generation.

5.2 Non-Functional Requirements

1. Reproducibility of setup and experiments.
2. Reasonable processing latency for telemetry and scoring.
3. Reliability under concurrent scenario execution.
4. Controlled access to orchestration endpoints.
5. Maintainable modular architecture.

5.3 Environment Requirements

Item	Requirement
Host	Virtualization-capable system
Firewall VM	OPNsense with multiple interfaces
Analytics VM	Ubuntu with log + model stack
IDS/IPS	Suricata integration
Data stack	Syslog + SIEM/dashboard tooling
Modeling	Python stack (pandas, scikit-learn, joblib)
Traffic generation	TRex + scripted scenarios

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6. Methodology

Method follows build-measure-improve cycle for network defense engineering.

6.1 Stage 1: Baseline Architecture and Policy

1. define trust boundaries and subnets,
2. configure explicit allow/deny policy,
3. validate route correctness and policy hit behavior.

6.2 Stage 2: Detection and Telemetry

1. enable Suricata and tune ruleset selection,
2. verify syslog forwarding and event integrity,
3. align timestamps for proper correlation.

6.3 Stage 3: Feature and Model Pipeline

1. transform firewall/IDS events into flow-level statistical features,
2. train Isolation Forest baseline on mostly normal traffic,
3. train Random Forest on labeled scenario dataset,
4. integrate outputs into dashboard stream.

6.4 Stage 4: Operational Validation

1. run scenario-based tests,
2. execute TRex stress profiles,
3. compare visibility and detection quality across modes.

<image of methodology pipeline describing stage-wise flow from policy setup to telemetry, model scoring, and validation feedback>

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7. Experimentation

7.1 Experiment Setup

1. segmented test network through OPNsense,
2. Suricata active on relevant interfaces,
3. centralized telemetry and dashboard,
4. model scoring pipeline attached to processed logs,
5. traffic from scripted tests and TRex profiles.

7.2 Experiment Categories

7.2.1 Policy Behavior Tests

1. allowed flow confirmation,
2. denied flow confirmation,
3. restricted egress confirmation.

7.2.2 Detection Tests

1. known-signature scenarios,
2. mixed traffic with overlapping patterns,
3. event correlation between firewall and IDS streams.

7.2.3 Model Tests

1. anomaly score distribution under normal traffic,
2. anomaly spikes under suspicious traffic,
3. classification quality for labeled attack classes.

7.2.4 Load Tests

1. low to high TRex profile ramp,
2. pipeline lag observation,
3. packet drop and resource usage capture.

<image of traffic load progression describing expected detector behavior from low, medium, and high profile intensity>

7.3 Observation Metrics

Metric Group	Metrics
Detection	event count, severity distribution, signature diversity
Model	confusion matrix, precision, recall, macro F1, anomaly score spread
System	CPU, RAM, ingestion delay, packet drop
Operations	run repeatability, scenario-to-alert correlation

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8. Results

This section is currently a working draft projection and will be updated with final measured values.

8.1 Expected Outcome Summary

1. clear policy enforcement between segmented zones,
2. improved visibility through unified telemetry,
3. anomaly layer helps spot suspicious behavior beyond direct signatures,
4. supervised model helps classify known scenario types for faster triage.

8.2 Reporting Template for Final Numbers

Area	Metric	Value
Signature detection	Total events	[ ]
Signature detection	Distinct signatures	[ ]
Signature detection	High/Critical share	[ ]
Anomaly model	Anomaly ratio	[ ]
Anomaly model	Estimated false positives	[ ]
Classification model	Accuracy	[ ]
Classification model	Macro F1	[ ]
Performance	Avg ingest delay	[ ]
Performance	Peak analytics CPU	[ ]
Performance	Packet drop at stress	[ ]

8.3 Discussion

From the literature and current implementation trend, some observations are expected:

1. signature-based engines stay strong for known threats,
2. anomaly scoring can surface unknown or less-obvious activity,
3. classification quality is strongly tied to label quality,
4. sidecar pattern keeps enforcement stable while analytics evolves.

8.4 Current Limitations

1. testbed traffic is smaller than enterprise traffic,
2. model behavior depends on feature quality and class balance,
3. encrypted traffic depth remains a challenge,
4. full auto-blocking needs strict guardrails to avoid wrong blocks.

8.5 Projection and Future Direction

1. include richer protocol metadata,
2. use dynamic thresholding by time and traffic profile,
3. evaluate long-window drift behavior,
4. build safer response-policy layer with review controls.

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9. Deliverables

9.1 Technical Deliverables

1. configured OPNsense segmentation and firewall policy,
2. Suricata-based IDS/IPS setup,
3. centralized logging and dashboard views,
4. anomaly and classification model scripts/artifacts,
5. scenario and TRex traffic profiles,
6. experiment logs and observations.

9.2 Documentation Deliverables

1. architecture write-up,
2. setup and configuration notes,
3. methodology and metric definitions,
4. complete project report.

9.3 Academic Deliverables

1. literature-based problem framing,
2. comparative network defense analysis,
3. conclusions and future work proposal.

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